

Tilted In-Hand Regrasping with Reinforcement Learning

Environment, control stack, reward design and learned policy
for the Franka regrasp task in Genesis, with the grasp axis tilted by $\theta \in [0^\circ, 45^\circ]$

Genesis regrasp project

24 September 2026

Summary

- **Task.** A 35 g, $25 \times 25 \times 150$ mm bar is held between the fingers. The policy has to move it 12.5 mm–22.5 mm along its own axis inside the hand. It does this by throwing: it moves the hand along the grasp axis and fires short gripper pulses. `env_franka_parallel_tilted.py` tilts that axis by a fixed θ up to 45° and keeps observations, reward and termination in a fixed task frame. At $\theta = 0$ it reproduces the original environment exactly. The unit tests confirm that observations match a rotated copy of the original for $\theta \in \{0, 15, 30, 45\}^\circ$.
- **Policy.** A GRU actor-critic trained with PPO. The best tilted run (`tilt45-gru-speed`, 45° , domain randomization on) ends 95.6 % of training episodes in success over its last 50 iterations, with mean episode length 3.7 s. Of the remaining episodes, 3.6 % fail, and 92 % of those failures are the seventh-regrasp limit.
- **Strategy.** The policy rises to the 0.45 m/s velocity cap, fires the pulse so the gripper opens as the command reverses at the 15 m/s^2 acceleration cap, and closes again while the hand moves down. The bar keeps its upward momentum and moves up the hand. Each throw moves it 10 mm–12 mm at 45° and costs several centimetres of hand height.
- **Physics.** A 1-D model of the release reproduces Genesis’ slip within 0.3 mm at 0° , and within 0.2 mm at 40° – 45° at the nominal friction (3 mm across $\mu \in [0.25, 1]$). It needs gravity along the axis *and* friction on the lower finger, because the transverse part of gravity loads the lower finger (Section 5). The model shows that 0.45 m/s is too slow for the policy’s 120 ms–160 ms release window. At 0.65 m/s Genesis gives 24 mm per throw, independent of window length.
- **Reward.** With $\gamma = 0.99$, the 1.2 s pulse lockout multiplies the discounted success reward by 0.55 for every extra pulse. So slip per pulse is the quantity that decides both episode length and return.

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1 Task and problem statement

The Franka Panda holds a 35 g bar ($25 \times 25 \times 150$ mm, MJCF friction 0.75) in a pinch grasp. Let r_z be the bar centre’s position relative to the midpoint of the two fingertips, measured along the grasp axis. At reset $r_z \approx 0$, and each episode draws a target $r_z^* \sim \mathcal{U}[12.5 \text{ mm}, 22.5 \text{ mm}]$ (a random sign with `-mix`, negative with `-negative`). The fingers never let go for good. The only way to change r_z is a short release while the hand accelerates, followed by a re-grasp: a *dynamic regrasp*.

The policy acts at 50 Hz ($T = 20$ ms). Its outputs are a velocity command along the grasp axis and a gripper action that can only *trigger* a fixed-shape pulse. Physics runs at 1 kHz. The tilted variant rotates the whole grasp, and therefore the direction of motion, by an angle θ away from vertical. The rotation is fixed for a run, and θ is not observed. Training in this report uses $\theta = 45^\circ$, the largest supported tilt.

2 The tilted environment

2.1 Home posture: inverting joint 5

The untilted home is $\mathbf{q}_0 = (0, -0.82, 0, -2.18, 0, 2.9, 0.78)$ rad, with the grasp axis vertical. Tilting is done with joint 5 alone. At home its axis $\hat{\mathbf{a}}$ is not horizontal ($a_z \neq 0$), so $q_5 = -\theta$ would give the wrong inclination. Rotating the vertical $\hat{\mathbf{e}}_z$ by q about $\hat{\mathbf{a}}$ (Rodrigues’ formula) gives

$$\hat{\mathbf{e}}_z \cdot R(\hat{\mathbf{a}}, q) \hat{\mathbf{e}}_z = a_z^2 + (1 - a_z^2) \cos q = \cos \theta \quad \implies \quad q_5(\theta) = -\arccos \frac{\cos \theta - a_z^2}{1 - a_z^2}. \quad (1)$$

$\hat{\mathbf{a}}$ is read from forward kinematics at start-up (`_configure_home_pose`). With the unit-test fixture $a_z = \cos 1.36 = 0.21$, this gives $q_5 = -15.3^\circ, -30.7^\circ, -46.1^\circ$ for $\theta = 15^\circ, 30^\circ, 45^\circ$. The test `test_joint_angle_produces_requested_inclination` checks the inversion to 10^{-7} degrees.

2.2 Fixed task frame

The frame is fixed at home and does not follow tracking errors. Let $\mathbf{q}_t = \mathbf{q}_{EE}(\theta) \otimes \mathbf{q}_{EE}(0)^{-1}$ be the change of hand orientation that the tilt produces. The task axes are the world axes rotated by it, $A = [\hat{\mathbf{x}}_t \ \hat{\mathbf{y}}_t \ \hat{\mathbf{z}}_t] = R(\mathbf{q}_t)$, so $A = I$ exactly at $\theta = 0$. With $\mathbf{p}_0$ the hand position at home and $Z_0 = \text{HOME_TASK_Z} = 0.8546$ m,

$$\mathbf{z}_t(\mathbf{p}) = Z_0 + \hat{\mathbf{z}}_t^\top (\mathbf{p} - \mathbf{p}_0), \quad \mathbf{v}_t = A^\top \mathbf{v}, \quad \mathbf{r} = A^\top (\mathbf{p}_{\text{bar}} - \tfrac{1}{2}(\mathbf{p}_{\text{tip},l} + \mathbf{p}_{\text{tip},r})). \quad (2)$$

Every observation, reward distance, success and failure bound, and hold target is expressed through (2). Gravity stays world-down, so in task coordinates it has an axial part $-g \cos \theta$ and a transverse part of magnitude $g \sin \theta$. At 45° Genesis gives $A^\top \mathbf{g} = (-0.61, 6.91, -6.94) \text{ m/s}^2$. The transverse part lies along the finger-closing axis (cosine -0.996), so the bar rests on the *lower* finger whenever the grip opens (Figure 1). The consequences are worked out in Section 5.

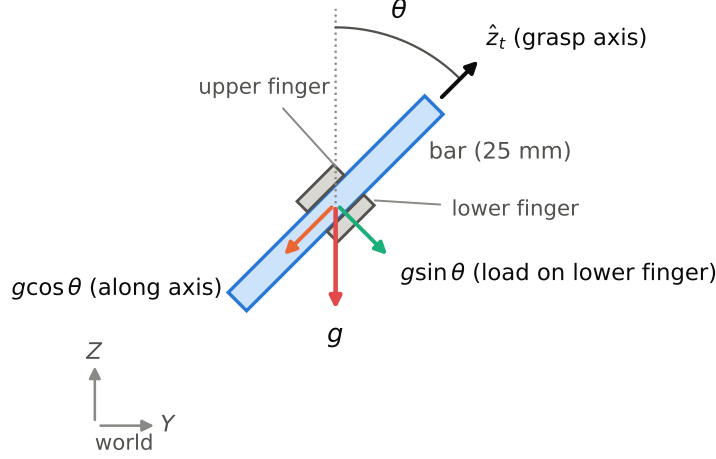


Figure 1: Tilted grasp in the plane of the tilt. Gravity splits into $g \cos \theta$ along the grasp axis and $g \sin \theta$ across it. In this posture the transverse part points along the finger axis, so an open grip leaves the bar resting on the lower finger.

2.3 Observation and action

The observation layout is identical to the untilted environment, so existing policies load unchanged. With `-normalization`, each channel is divided by the fixed scale below. With `-randomize`, $\mathcal{U}(\pm 1 \text{ mm})$ noise (or $\pm 1 \text{ mm/s}$) is added to channels 0, 1, 6, 7 and 8.

idx	symbol	definition	scale
0	z_t	hand position along the axis, Eq. (2)	1.0
1	$\dot{z}_t$	hand velocity along the axis, $\dot{z}_t^\top \mathbf{v}_{EE}$	0.6
2, 3	v_k, a_k	current velocity reference and its step acceleration	0.6, 15
4, 5	$ \mathbf{F}_l , \mathbf{F}_r $	net contact-force magnitude on each finger	5, 5
6, 7, 8	r_z, r_x, r_y	bar relative to the fingertip midpoint, task frame	0.05
9	r_z^*	target in-hand offset	0.05

The action is $\mathbf{a} \in [-1, 1]^3$. The first component sets the velocity reference and the other two set the finger targets $p_f = p_{\min} + \frac{1}{2}(a + 1)(p_{\max} - p_{\min})$, with $p_{\min} = 0.251 \text{ mm}$ and $p_{\max} = 13.0 \text{ mm}$ per finger. The open gap of 26 mm clears the bar by 0.5 mm per side. Outside a pulse the fingers are clamped to $p_{\min}$, so the finger action serves only as a pulse trigger (Section 2.5).

2.4 From action to motion

Reference. The velocity command is scaled and then rate-limited:

$$v_k = \text{clip}(a_0 v_{\max}, v_{k-1} \pm a_{\max} T), \quad v_{\max} = 0.45 \text{ m/s}, \quad a_{\max} = 15 \text{ m/s}^2. \quad (3)$$

The position knot is integrated with the trapezoid rule, and the step acceleration is the finite difference:

$$z_k = z_{k-1} + \frac{1}{2}(v_{k-1} + v_k)T, \quad a_k = (v_k - v_{k-1})/T. \quad (4)$$

At 1 kHz the reference is the cubic Hermite spline through (z_{k-1}, v_{k-1}) and (z_k, v_k) , $z(s) = h_{00}z_{k-1} + h_{10}Tv_{k-1} + h_{01}z_k + h_{11}Tv_k$ with $s \in [0, 1]$.

Lemma. With trapezoid-consistent knots, the Hermite velocity is exactly the linear interpolation of the knot velocities, and its acceleration is exactly a_k :

$$\dot{z}(s) = (1 - s)v_{k-1} + sv_k, \quad \ddot{z} = a_k. \quad (5)$$

Proof. Differentiate: $T\dot{z} = h'_{00}z_{k-1} + h'_{10}Tv_{k-1} + h'_{01}z_k + h'_{11}Tv_k$. Using $h'_{01} = -h'_{00} = 6s - 6s^2$ and substituting $z_k - z_{k-1} = \frac{1}{2}(v_{k-1} + v_k)T$, the s^2 terms cancel and the rest collects to $T[(1 - s)v_{k-1} + sv_k]$. $\square$

So the observed a_k is the true reference acceleration. The reference jerk is impulsive at every knot, and that is what the command filter smooths.

Command filter. The velocity reference passes through $H(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$, with $\omega_n = 75$ rad/s and $\zeta = 0.33$, tuned against the real arm. It is discretized at $T_s = 1$ ms with the Tustin substitution $s \mapsto \kappa \frac{1-z^{-1}}{1+z^{-1}}$, $\kappa = 2/T_s$:

$$\alpha_0 = \kappa^2 + 2\zeta\omega_n\kappa + \omega_n^2, \quad b_0 = b_2 = \frac{\omega_n^2}{\alpha_0}, \quad b_1 = \frac{2\omega_n^2}{\alpha_0}, \quad a_1 = \frac{2\omega_n^2 - 2\kappa^2}{\alpha_0}, \quad a_2 = \frac{\kappa^2 - 2\zeta\omega_n\kappa + \omega_n^2}{\alpha_0}. \quad (6)$$

Its low-frequency delay is $2\zeta/\omega_n = 8.8$ ms and its step overshoot is $e^{-\pi\zeta/\sqrt{1-\zeta^2}} = 33\%$. The overshoot matters for the throw: a commanded $+0.45 \rightarrow -0.45$ reversal becomes a hand swing of about $+0.47 \rightarrow -0.61$ m/s (Figure 2b).

Cartesian controller. This runs at 1 kHz. With $e = p^{\text{ref}} - p_{\text{EE}}$ and the axial projector $P = \hat{z}_t \hat{z}_t^\top$, the commanded twist is

$$v^{\text{cmd}} = \hat{z}_t \tilde{v} + K_{\text{ax}} P e + K_{\text{tr}} (I - P) e, \quad \omega^{\text{cmd}} = K_r \text{rotvec}(q^{\text{ref}} \otimes q_{\text{EE}}^{-1}). \quad (7)$$

Here $\tilde{v}$ is the filtered reference, $K_{\text{ax}} = 8$, $K_{\text{tr}} = 80$ and $K_r = 40 \text{ s}^{-1}$. It is mapped to joint velocities with damped least squares, $\dot{q} = J^\top (J J^\top + \lambda I)^{-1} [v^{\text{cmd}}; \omega^{\text{cmd}}]$ with $\lambda = 10^{-4}$, and tracked by the joint velocity loop ($k_v = 0.35 \times$ nominal, torque limits 87 Nm/12 Nm). The transverse gain is ten times the axial gain so the hand stays on the tilted line against the gravity-induced sideways drift. The axial loop is unchanged from the untilted environment. The unit test `test_control_holds_orientation_and_commands_axis_velocity` checks that a lateral offset and a yaw error produce restoring commands without changing the axial speed.

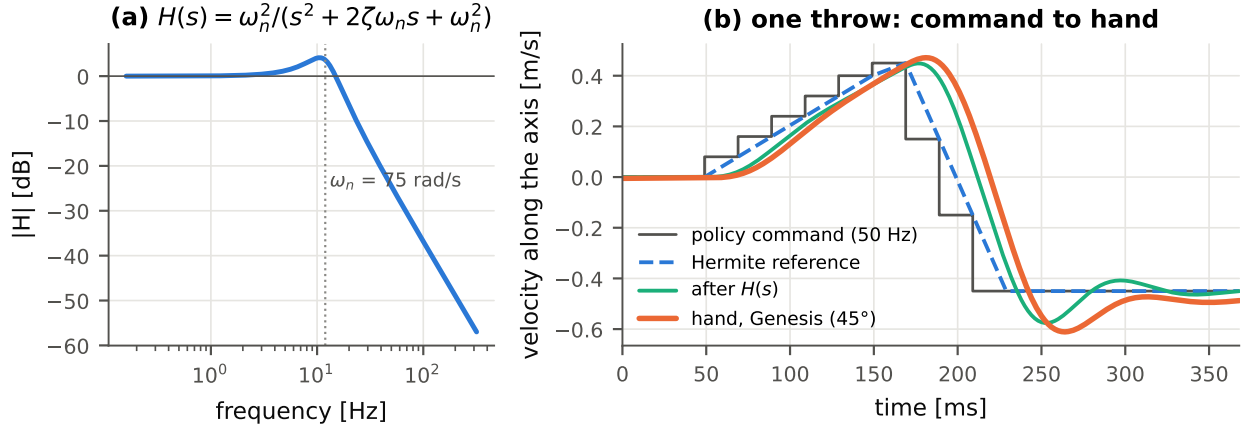


Figure 2: (a) Magnitude of the command filter. (b) One throw at 45° : the 50 Hz policy command, the Hermite reference (a linear ramp inside each step, by the Lemma), the filtered reference, and the hand velocity Genesis measures. The hand overshoots the -0.45 m/s command to -0.61 m/s.

2.5 Gripper pulse

A pulse fires on a rising edge of the *effective* mean finger command through the mid-position. Pulses are blocked during the first 50 steps (1 s) and for $N_\ell = 60$ steps (1.2 s) after the previous trigger. On a trigger the counter is set to $c = L + D$, with $D = 2$ and $L = 7$. For $c > L$ the policy’s command passes through (this is the delay). For $2 \leq c \leq L$ the fingers are forced open for $L - 1 = 6$ steps (120 ms). At $c = 1$ they are forced closed. Afterwards they are clamped closed. If the policy keeps commanding “open” through the delay, the bar can be loose for up to $(D + L - 1)T = 160$ ms. Because the edge detector remembers the effective command, holding “open” through the lockout still produces an edge the moment the lockout ends. Figure 3 shows the pulse together with the reward windows keyed to the trigger.

2.6 Termination and randomization

Success requires all of the following for 5 consecutive steps: $|r_z - r_z^*| \leq 5$ mm, $|\dot{z}_t| < 0.05$ m/s, a firm grasp (mean finger force $\bar{F} \geq 0.75$ N) for at least 3 steps, and $z_t \in [0.70, 0.86]$ m. **Failure** is any of $|r_x|$ or $|r_y| > 40$ mm, fingertip distance < 10 mm, $|r_z| > 150$ mm, $z_t \notin [0.60, 0.96]$ m, or a seventh regrasp. **Timeout** is 450 steps (9 s). Every reset samples the finger gain $k_p \sim \mathcal{U}[100, 500]$ (force limit $= k_p$) and the contact friction $\mu \sim \mathcal{U}[0.6, 0.9]$, whether or not `-randomize` is set. `-randomize` adds the observation noise above and initial joint-velocity noise of ± 0.01 rad/s. The pulse delay is fixed at two steps in every mode.

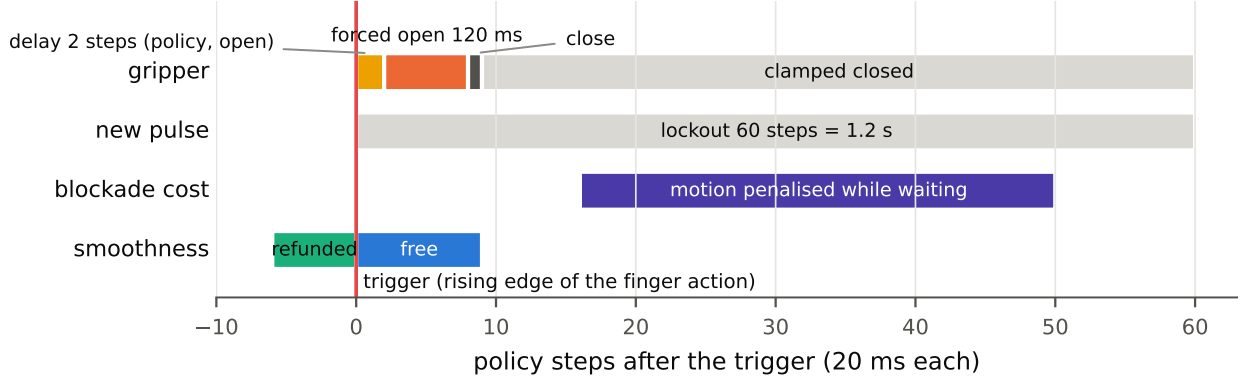


Figure 3: Pulse state machine and the reward windows attached to the trigger (current constants). The smoothness refund and exemption are defined in Section 3.4, the blockade window in Section 3.5.

3 Reward

The per-step reward is a weighted sum. Only the keys of `REWARD_TERM_WEIGHTS` are summed:

$$r = r_{\text{base}} + r_{\text{prox}} + 2r_{\text{regrasp}} + r_{\text{acc}} + r_{\text{jerk}} + r_{\text{flip}} + r_{\text{blockade}} + r_{\text{speed}}, \quad (8)$$

with $r_{\text{flip}} \equiv 0$ at present, because its coefficient is 0. The post-pulse hold penalty is computed for its countdowns but is not summed. Figure 4 plots each term.

3.1 Terminal and base reward

$$r_{\text{base}} = \begin{cases} R_s - c_t k & \text{success at step } k, \quad R_s = 10^4, \quad c_t = 0.5 \\ -F & \text{failure} \\ -R_{\text{to}} & \text{timeout, } R_{\text{to}} = 250 \\ -c_a & \text{otherwise, } c_a = 1.25. \end{cases} \quad (9)$$

Undiscounted, failing at step t returns $-c_a(t-1) - F$ and timing out returns $-c_a(L_{\text{max}}-1) - R_{\text{to}}$. Failing is strictly worse for every $t \geq 1$ if and only if $F > R_{\text{to}} + c_a(L_{\text{max}}-1)$. The code uses $F = R_{\text{to}} + c_a L_{\text{max}} + 100 = 912.5$, derived in `fail_penalty()`, so the ordering survives any change to the constants.

3.2 Proximity

$$r_{\text{prox}}(d) = M \frac{e^{-kd/R} - e^{-k}}{1 - e^{-k}}, \quad d = |r_z - r_z^*|, \quad M = 30, \quad R = 50 \text{ mm}, \quad k = 5, \quad (10)$$

clipped to $[0, M]$. It pays 30, 18.1, 10.9 and 3.9 at $d = 0, 5, 10$ and 20 mm .¹ It is dense but bounded, and Section 3.6 shows that loitering cannot beat finishing.

3.3 Regrasp bonus

A *release* starts when $\bar{F} < 0.15 \text{ N}$. A *regrasp event* is the first step after a release with $\bar{F} \geq 0.75 \text{ N}$. With e_b and e_a the target errors $|r_z - r_z^*|$ at the release and at the regrasp, let $\delta = e_b - e_a$. Then

$$\rho(\delta) = 7500 \max(\delta, -0.05) \left(\frac{|\delta|}{7.5 \text{ mm}} \right)^3 \text{ clipped above at } 10^4, \quad r_{\text{regrasp}} = \begin{cases} \rho & \rho \geq 0 \\ 5\rho & \rho < 0, \end{cases} \quad (11)$$

paid only on the first four regrasps. The bonus is *quartic*, $\propto \delta|\delta|^3$. Weighted by 2, it pays 112.5, 356, 1800 and 5689 for improvements of 7.5, 10, 15 and 20 mm. One 20 mm regrasp therefore earns eight times as much as two 10 mm regrasps. Since $\delta \leq e_b \leq 22.5 \text{ mm}$ in a normal episode, the weighted bonus stays below 9.1×10^3 and the cap (reached at $\delta = 27.4 \text{ mm}$) is not binding. A worsening costs five times the bonus of the same improvement, so a deliberate “worsen then improve” cycle loses reward.

¹The comment above `PROXIMITY_REWARD_MAX` still quotes the values for $M = 20$.

3.4 Smoothness with pulse refunds

$$r_{\text{acc}} = -25 \left(\frac{a_k}{a_{\text{max}}} \right)^2, \quad r_{\text{jerk}} = -25 \left(\frac{a_k - a_{k-1}}{2a_{\text{max}}} \right)^2. \quad (12)$$

Both are at most 25 per step: a full-scale alternating acceleration costs 50 per step, or 1500 over 30 steps. The throw needs exactly such accelerations, so both terms have a refund mechanism. On an *accepted* trigger:

- the charges of the previous six steps are paid back once;
- the trigger step and the next eight steps are free.

The run-up and the reversal are therefore free exactly when they lead to a real pulse, and cost full price otherwise. The refund cancels the charges exactly in undiscounted terms. Under discounting a charge j steps before the refund keeps a residue $1 - \gamma^j \leq 1 - 0.99^6 = 5.9\%$. Five unit tests pin the bookkeeping: the window boundaries, no double refund, per-environment resets, and that a trigger rejected by the lockout refunds nothing.

3.5 Speed and blockade

With the measured speed above a 0.02 m/s deadzone, $u = \max(|\dot{z}_t| - 0.02, 0)$,

$$r_{\text{speed}} = -2.25 \left(\frac{u}{v_{\text{max}}} \right)^2, \quad r_{\text{blockade}} = -\mathcal{K}_{\text{idle}} \left[\left(\frac{u}{0.1} \right)^2 + 0.25 \left(\frac{v_k}{0.1} \right)^2 \right]. \quad (13)$$

$\mathcal{K}_{\text{idle}}$ is 1 while no pulse is running, at least 0.3 s have passed since the trigger (settle), and more than 0.2 s of the block remains (prep). That is steps 16–50 of each lockout, and steps 1–40 of an episode (Figure 3). Motion is never free, but it is most expensive while the policy waits for the next pulse.

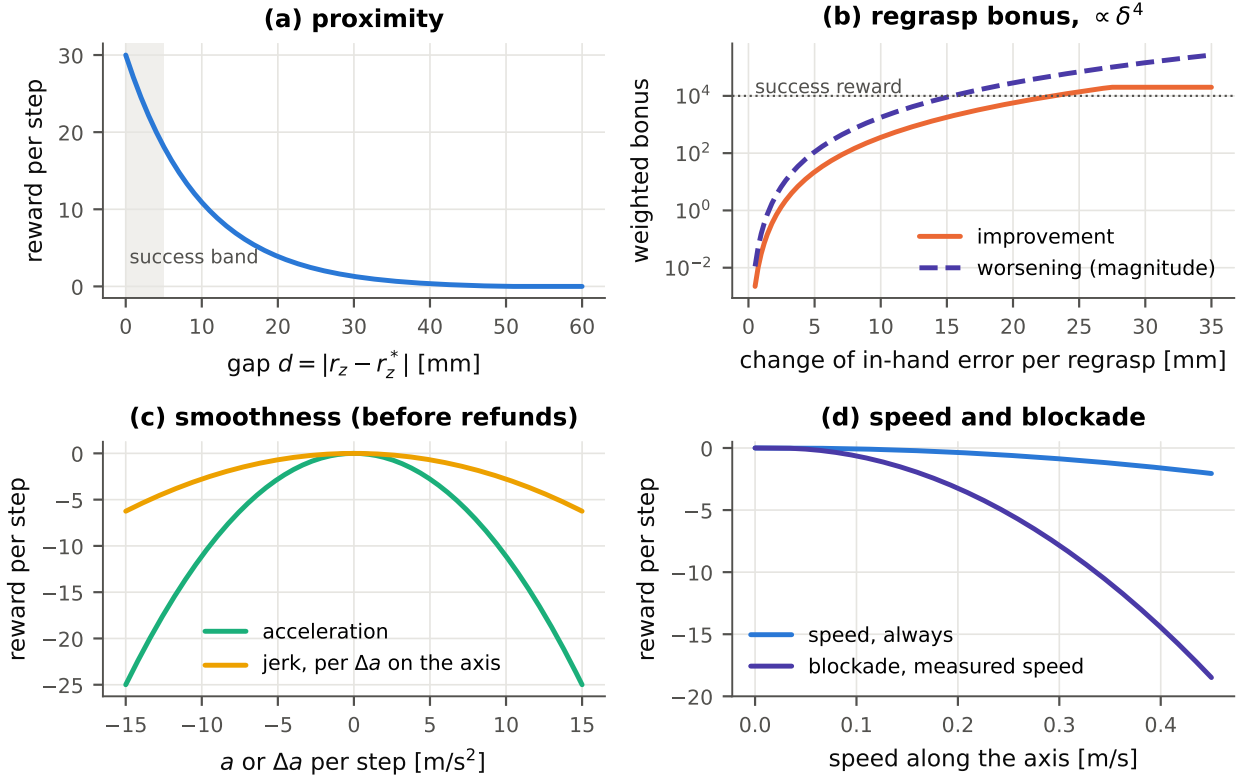


Figure 4: Reward terms with the current constants: (a) proximity; (b) weighted regrasp bonus on a log scale, where improvement and worsening are $\propto \delta^4$ and worsening is five times heavier; (c) smoothness before refunds; (d) speed and blockade penalties.

3.6 What the discount does to the design

PPO optimizes $\sum_k \gamma^k r_k$ with $\gamma = 0.99$, an effective horizon of $1/(1 - \gamma) = 100$ steps (2s). The earliest success after n pulses comes at step $t_n \approx 50 + 60(n - 1) + 12$: the start block, then one lockout per extra

pulse, then settling. The discounted terminal value falls by $\gamma^{60} = 0.547$ per pulse. Figure 5 evaluates the base stream exactly. Finishing after 1, 2, 3, 4 or 5 pulses is worth 5343, 2858, 1502, 763 and 359. Loitering to timeout at 6 mm from the target is worth 1493 (just outside the success band), and at 12 mm it is worth 753.

- **A policy that needs four or more pulses gains nothing over hovering near the target,** before the regrasp bonus is counted. The bonus, quartic in slip per pulse, points the same way.
- **Slip per pulse is therefore the quantity that sets the return.** Section 5 shows that it is limited by the 0.45 m/s velocity cap, not by the policy.

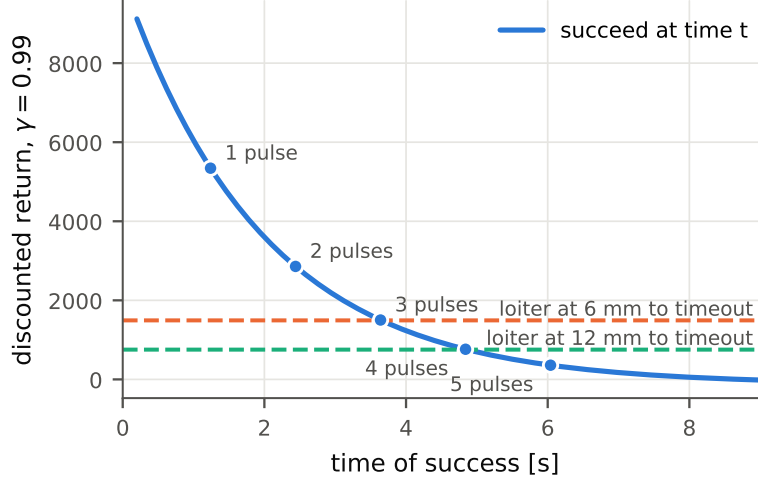


Figure 5: Discounted base return against the time of success (alive cost included), with the earliest success after n pulses marked. The dashed lines are loitering at a fixed distance until timeout.

4 Learning

4.1 PPO with a recurrent actor-critic

Training uses rsl_rl 5 PPO. With $\rho_t = \pi_\theta(a_t|h_t)/\pi_{\text{old}}(a_t|h_t)$ and GAE advantages

$$\hat{A}_t = \sum_{l \geq 0} (\gamma \lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(h_{t+1}) - V(h_t), \quad (14)$$

the minimized loss is

$$\mathcal{L} = \mathbb{E}[\max(-\rho_t \hat{A}_t, -\text{clip}(\rho_t, 1 \pm \epsilon) \hat{A}_t)] + c_V \mathbb{E}[\max((V - R)^2, (V_{\text{old}} + \text{clip}(V - V_{\text{old}}, \pm \epsilon) - R)^2)] - c_H \mathbb{E}[\mathcal{H}(\pi_\theta)]. \quad (15)$$

The learning rate adapts to the measured KL divergence. It is divided by 1.5 (floor 10^{-5}) when $\text{KL} > 2\text{KL}^*$ and multiplied by 1.5 (cap 10^{-2}) when $\text{KL} < \text{KL}^*/2$.

γ, λ	0.99, 0.95	clip ϵ	0.2
KL^* , initial lr	0.01, 3×10^{-4}	c_V, c_H	1.0, 0.005
epochs $\times$ minibatches	5×4	max grad norm	1.0
actor	GRU(256) $\rightarrow$ MLP 256-128-64, ELU	action std	scalar, init 1.0
critic	GRU(256) $\rightarrow$ MLP 256-128-64	rollout = BPTT horizon	64 steps (1.28 s)

Why recurrent. The observation contains no pulse phase, no lockout counter, no friction and no finger gain, although all of them change what a good action is. The GRU state $h_t = \text{GRU}(o_t, h_{t-1})$ can recover them from history. For example, the slip the last pulse produced carries information about μ . The 1.28 s BPTT window is just longer than one pulse cycle (60 steps), so the gradient can connect a trigger to the regrasp it caused. Recurrent minibatches split environments rather than time, so each minibatch holds whole 64-step sequences.

4.2 Runs

All runs below use $\theta = 45^\circ$, `-normalization`, `-randomize` and positive targets.

run	iterations	success, last 50 it.	best success (20-it. mean)	episode length
tilt45-gru	825	64.6 %	87.6 %	5.6 s
tilt45-gru-jerk	1014	77.4 %	86.7 %	4.2 s
tilt45-gru-speed	709	95.6 %	96.4 %	3.7 s
tilt45-gru-speed-only-up	418	82.5 %	85.8 %	2.1 s

These are training-time rates from the reward table, which uses the stochastic policy with randomization on. The run directories store no source snapshot, so the runs record successive revisions of the reward (named at training time) rather than a controlled ablation. The steps in Figure 6 coincide with resumes: iteration 700 for `tilt45-gru`, 600 for `-jerk` and 620 for `-speed`. Where a resume lowered the success rate, the curve settles at the new level rather than recovering. `tilt45-gru-speed` is the only run above 90 %. Its residual failures are almost entirely the seventh-regrasp limit (Figure 7): the policy runs out of attempts, not out of stability. `-speed-only-up` has the shortest episodes (2.1s), consistent with the discount argument of Section 3.6.

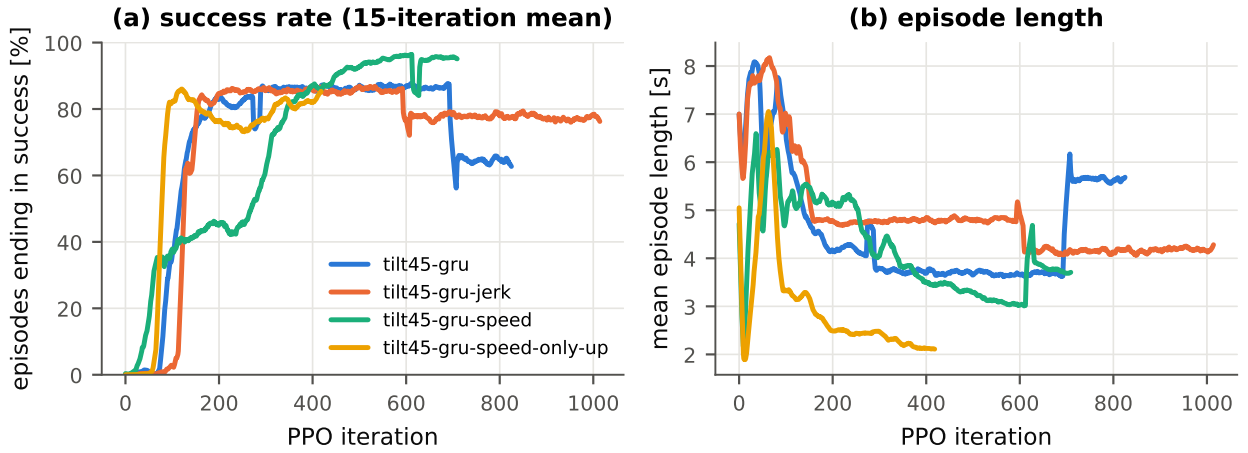


Figure 6: Training curves of the four 45° GRU runs (15-iteration moving mean).

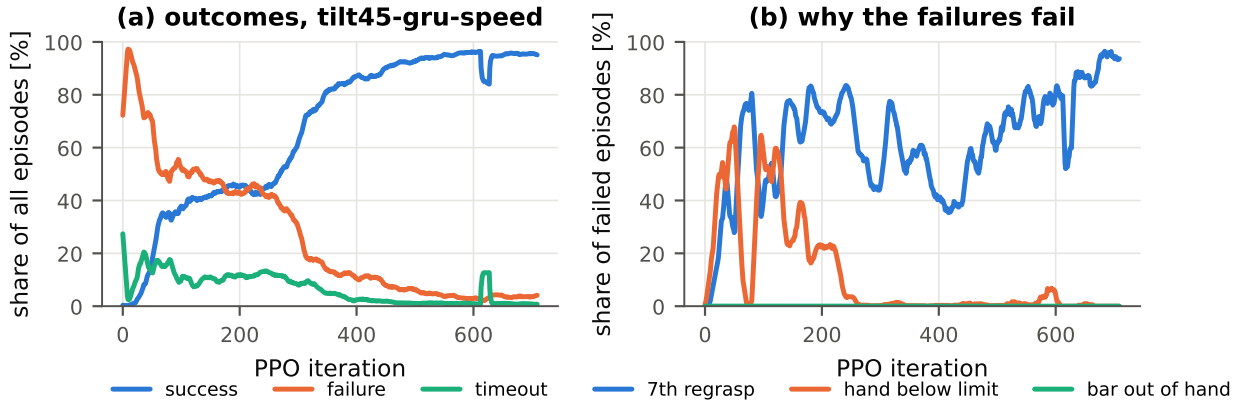


Figure 7: `tilt45-gru-speed`. (a) Outcomes as a share of all episodes. (b) Failure reasons as a share of failed episodes. The early failures are dominated by the hand leaving its height band; the late ones by the seventh-regrasp limit.

5 The learned strategy and its physics

5.1 The throw

Figure 8 shows the first 1.36s of the 45° episode exported for the real cell (`franka_controllers/traj_vis/tilt45/traj_d2_15.csv`, `tilt45-gru`, checkpoint 820). The policy drifts down slowly during the start

block. It then rises to the 0.45 m/s cap and fires so that the open window starts at the step where the command reverses at the -15 m/s^2 cap. It re-grips at -0.45 m/s . The bar gains 11.2 mm in one throw, and the hand ends about 50 mm lower along the axis. After the throw the command alternates at $\pm 15\text{ m/s}^2$ every step. This checkpoint predates the smoothness revision, and that chatter is what the acceleration and jerk costs of later runs target. Figure 9 is an episode of `tilt45-gru-speed-only-up` with four throws. The bar climbs $0 \rightarrow 6 \rightarrow 15 \rightarrow 18 \rightarrow 19\text{ mm}$ while the hand’s lowest point falls from 0.67 m to 0.60 m . The episode fails on the lower height limit. **Every throw spends hand height, so the height band caps the number of throws at about four**, independently of the regrasp limit.

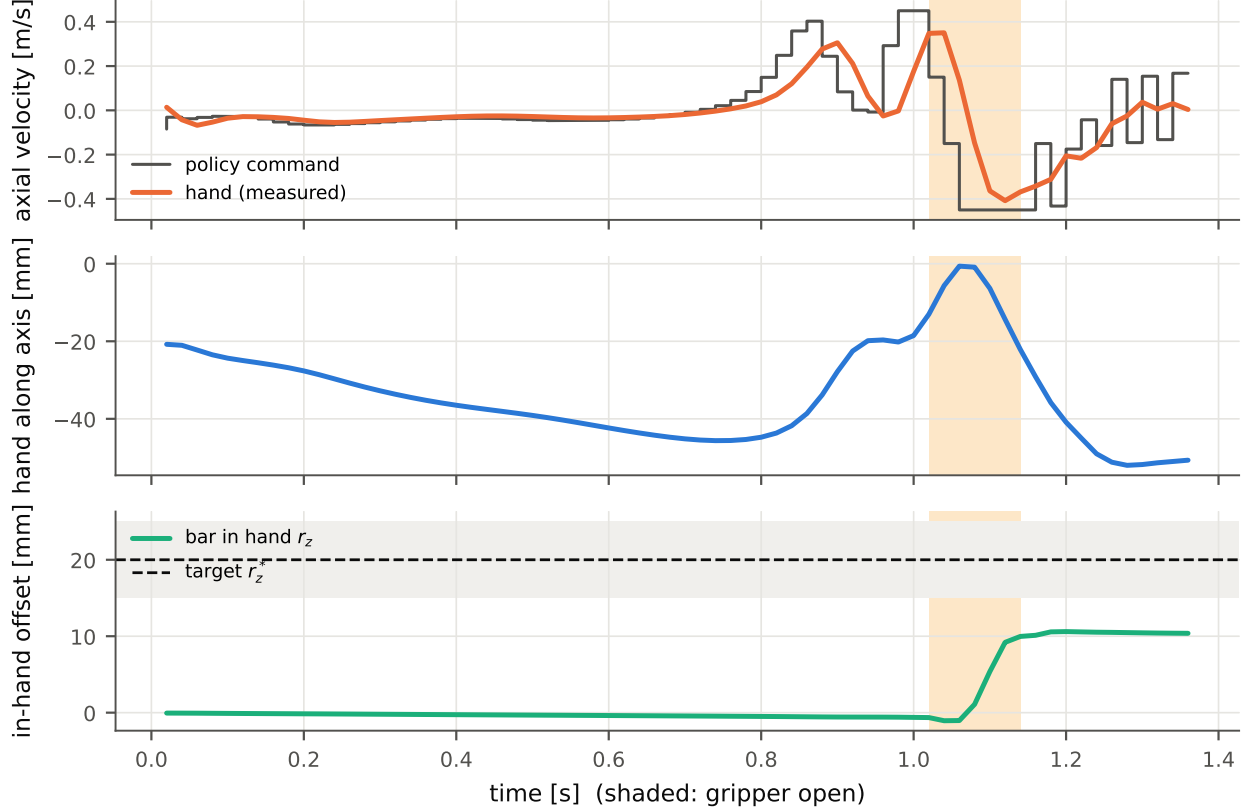


Figure 8: First 1.36s of the exported 45° trajectory; shading marks the open gripper. Top: policy command and measured hand velocity along the axis. Middle: hand position along the axis. Bottom: in-hand offset, with the target and its $\pm 5\text{ mm}$ success band.

5.2 Why the bar moves: relative dynamics

In the hand frame, with $w = v_{\text{bar}} - v_{\text{hand}}$ and slip $\Delta = \int w dt$ after a release, the bar rests on the lower finger (Figure 1). Its motion relative to the hand is

$$\dot{w} = \begin{cases} -g(\cos \theta + \mu \sin \theta) - a_h & w > 0 \\ -g(\cos \theta - \mu \sin \theta) - a_h & w < 0 \\ 0 & w = 0, |g \cos \theta + a_h| \leq \mu g \sin \theta. \end{cases} \quad (16)$$

The bar climbs the hand only while the hand decelerates faster than $g_+ = g(\cos \theta + \mu \sin \theta)$, which is 12.1 m/s^2 at 45° with $\mu = 0.75$. That explains why the policy uses its full 15 m/s^2 and why the filter overshoot helps it. Integrated with Genesis’ measured hand velocity, Eq. (16) reproduces the 1 kHz slip trace (Figure 10). Gravity along the axis alone would predict 29 mm instead of the 12 mm observed.

5.3 What limits the slip per throw

Figure 11 scripts the throw in Genesis across peak speed and open window. At $v = 0.45\text{ m/s}$ the best slip is 11 mm – 13 mm at every tilt, reached with a 100 ms – 120 ms window. The policy’s window is 120 ms forced and up to 160 ms loose, so it sits at or past that optimum. The companion report (*Pulse Duration*

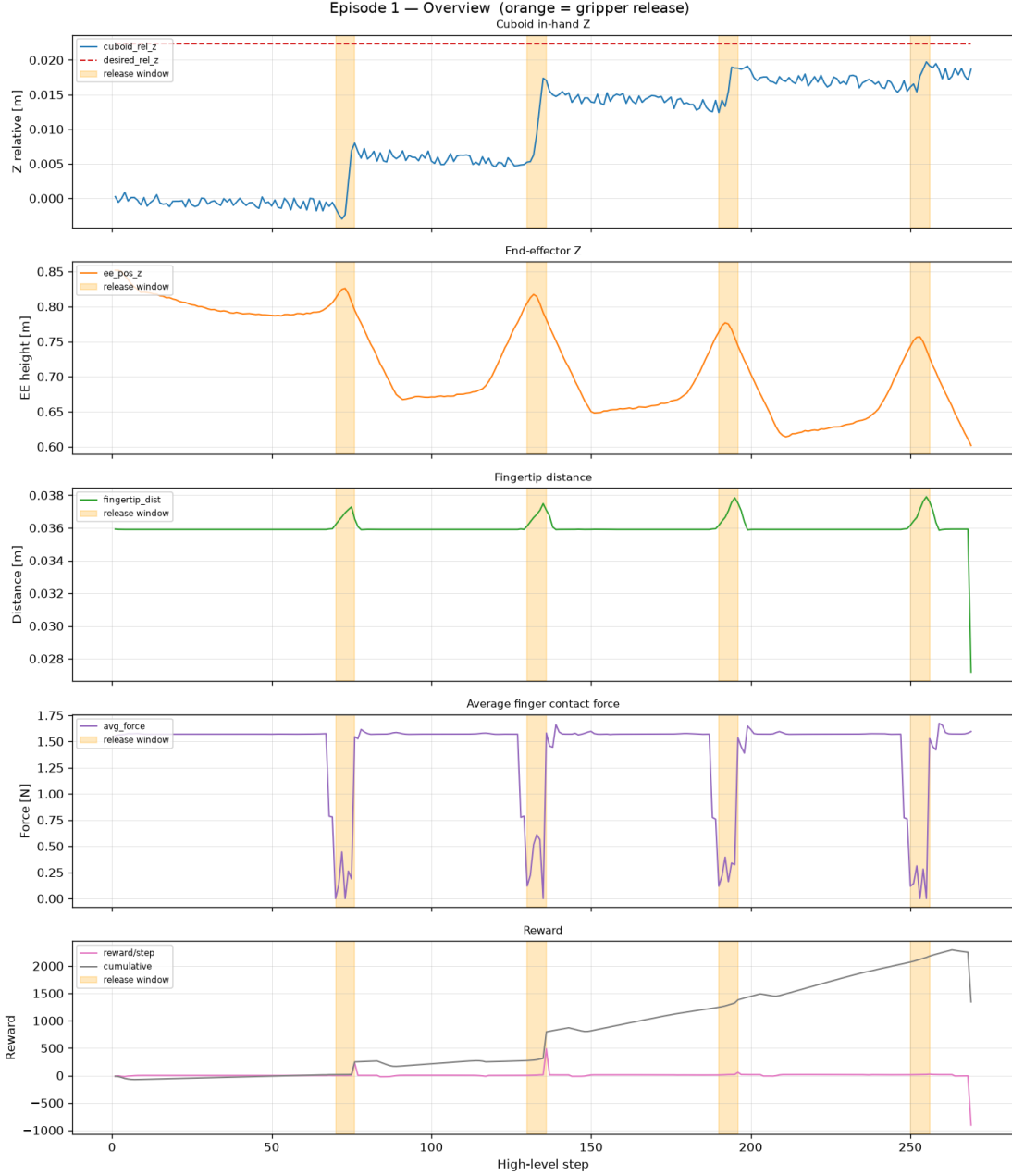


Figure 9: An episode of `tilt45-gru-speed-only-up`, plotted by the project’s evaluation tooling. Four throws move the bar up in steps of 3 mm–9 mm while the hand’s lowest point sinks with each throw, until $z_t < 0.60$ m ends the episode as a failure.

versus Arm Velocity) derives the matched condition $v^* = \frac{1}{2}g_+T_p$. For a 120 ms–160 ms window it asks for 0.6 m/s–0.8 m/s. With the cap lifted to 0.65 m/s, Genesis gives 24 mm–25 mm per throw at 45° for windows of 120 to 160 ms. That is more than the largest target (22.5 mm), so any target is reachable in one throw, with the peak speed as the knob that scales it down.

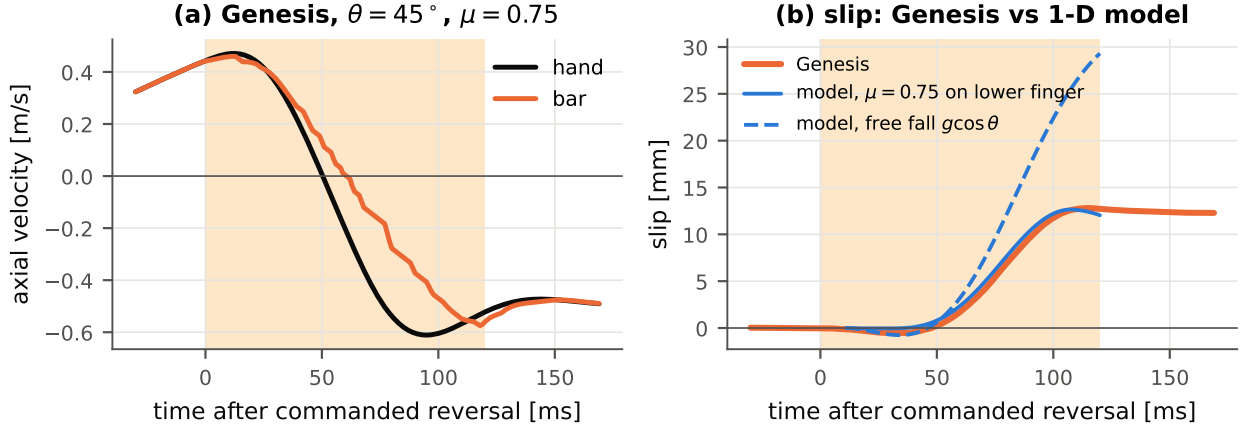


Figure 10: One scripted throw in Genesis at 45° , $\mu = 0.75$, logged at 1 kHz. (a) Hand and bar velocities along the axis; shading marks the forced-open window. (b) Slip against the 1-D model (16) driven by the measured hand velocity.

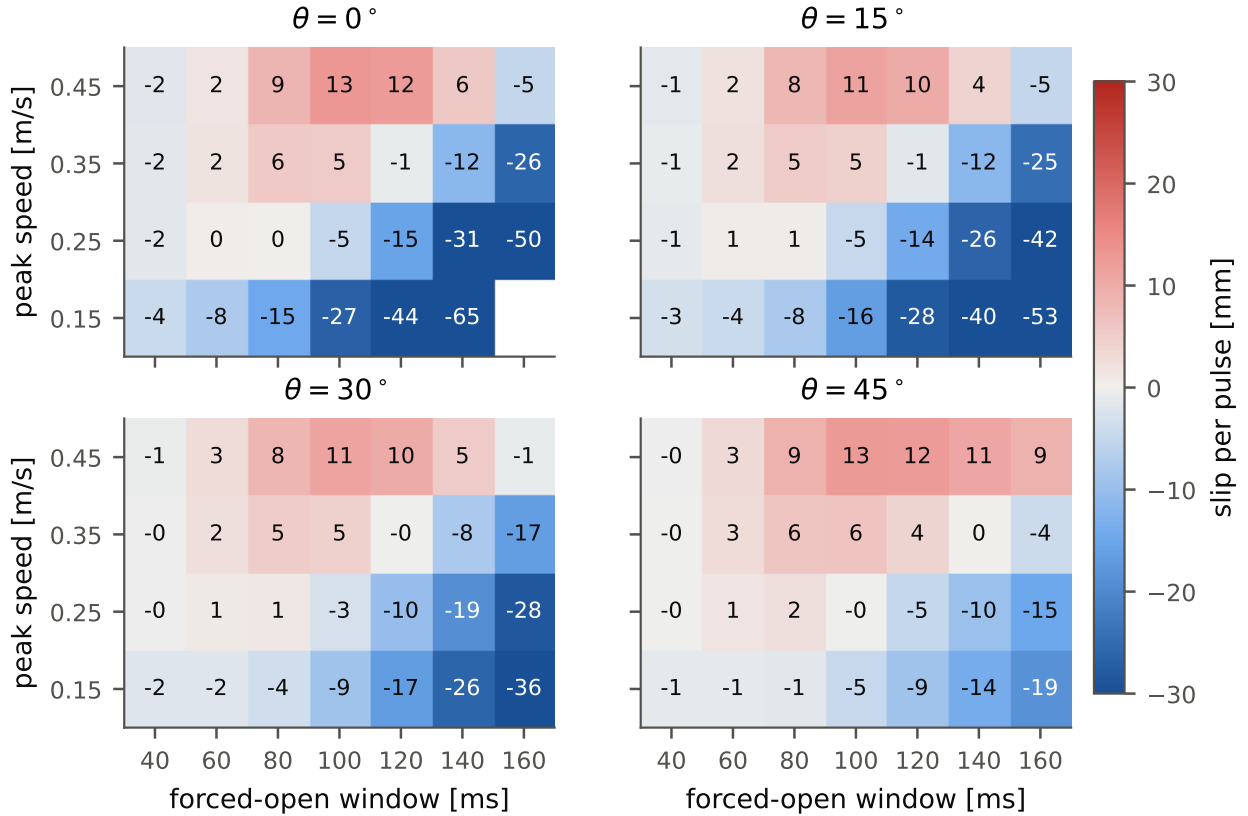


Figure 11: Genesis, scripted throws: slip per pulse [mm] against peak speed and forced-open window, trigger at the commanded reversal. Blank cell: that environment reset before re-grip.

6 Transfer to the real FR3

The learned behaviour reaches the robot as an open-loop replay of policy trajectories. Four pieces in `franka_controllers` make the real cell reproduce the simulated start state and motion:

- **Homing** (`home_fr3v2.py` `-tilt-deg`, `tilted_homing.py`). Joints 1–4 take the simulation values. Joints 5–7 are solved on the real URDF and gripper mounting so that the grasp direction matches. A hand-calibrated pose is used at 30° .
- **The simulation’s reset and the datum of a CSV** (`sim_home_tilted.py`). The simulator integrates from $v(0) = 0$ with the trapezoid rule, so the pose to home to is $z_0 = \text{target_z}[0] - \frac{1}{2}\text{target_z_vel}[0]T$. For the exported 45° episode this equals Z_0 to within 2×10^{-7} m. The script also checks the travel

envelope against joint limits with the controller’s own damped least squares.

- **Axis-following control** (`motion_mode: fixed_axis, ~/axis_vel_cmd`). This is the same structure as the simulator’s controller: the scalar speed is rotated onto a fixed line, with transverse position feedback and the orientation held at activation.
- **Replay** (`replay_traj_csv.py`) streams `target_z_vel` at 50 Hz and fires one Robotiq pulse per open window.

What does not transfer automatically.

- The gripper is different: a Robotiq 2F on the robot, the Franka hand model in simulation. Its latency and effective open time are not yet measured.
- The real reference chain adds one command period of interpolation lag. The reversal at 112.5 ms of the test profile falls between publish ticks.
- The contact friction of the real bar on the real pads is unknown, and Eq. (16) shows the slip depends on it strongly.

The companion report quantifies all three and gives the measurements that close them. Closed-loop deployment of the policy would also need the in-hand offset r_z on the robot, for example from the motion-capture setup in `hrii_optitrack`.

7 Recommendations

1. **Raise Z_VEL_MAX to about 0.65 m/s, or shorten the open window to about 80 ms.** Either choice matches the throw to the pulse (Section 5). The first gives about 24 mm per throw in Genesis, enough to finish in one pulse and worth roughly $1.9\times$ the discounted return of a two-pulse success (5343 vs 2858). Check the height budget first: a faster throw travels further.
2. **Consider rescaling the regrasp bonus.** The quartic bonus reaches 9.1×10^3 at the largest target, close to the success reward. A policy that becomes able to make large single throws will be driven mostly by the bonus.
3. **Fix the stale test and comment.** Five tests in `test_env_franka_parallel_tilted.py` target a `progress_reward` term that the environment does not define (it uses `proximity_reward`). They fail now: 12 of 17 pass. The proximity comment still quotes the $M = 20$ values.
4. **Observe the tilt when training across angles.** θ is fixed per run and absent from the observation. A single multi-tilt policy would need it, or would need to infer it through the GRU.

A Constants

policy / sim period	20 ms / 1 ms	episode	450 steps (9 s)
$v_{\max}, a_{\max}$	0.45 m/s, 15 m/s ²	filter ω_n, ζ	75 rad/s, 0.33
$K_{\text{ax}}, K_{\text{tr}}, K_r, \lambda$	8, 80, 40, 10^{-4}	joint k_p	4500/3500/2000
fingers closed / open	0.251 / 13.0 mm	finger k_p, k_v	$\mathcal{U}[100, 500]$, 25
pulse D, L	2, 7 steps	lockout, start block	60, 50 steps
release / firm force	0.15 / 0.75 N	regrasp limit	fail on 7th; bonus on first 4
$R_s, c_t, c_a, R_{\text{to}}, F$	10^4 , 0.5, 1.25, 250, 912.5	proximity M, R, k	30, 50 mm, 5
smoothness weights	25, 25; refund 6, free 9	speed, blockade	2.25; settle 0.3, prep 0.2 s
friction μ	$\mathcal{U}[0.6, 0.9]$	target r_z^*	$\mathcal{U}[12.5, 22.5]$ mm

B Reproduction

```
# train (genesis container)
python3 examples/rigid/train_franka_ppo_gru.py --env env_franka_parallel_tilted \
    --tilt-deg 45 -e tilt45-gru-speed --normalization --randomize -B 1024
# unit tests (no scene is built)
cd examples/rigid && python3 -m unittest test_env_franka_parallel_tilted
# Genesis throw experiments, then figures and this PDF
python3 reports/tools/pulse_slip_sweep.py --tilts 0 15 30 45 \
    --out reports/data/pulse_slip.npz
```

```
python3 reports/tools/pulse_hires.py --tilts 0 40 45 \  
    --out reports/data/pulse_hires.npz  
python3 reports/rl_tilted/make_figs.py  
cd reports/rl_tilted && latexmk -pdf rl_tilted_regrasp_report.tex
```

Learning curves are read from the runs' TensorBoard files into `reports/data/tb_curves.npz`. All quoted numbers are written to `reports/rl_tilted/numbers.json`.